

NUTRITION LABEL

FINAL PRESENTATION

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DATA COLLECTION

- ▶ Instagram posts
- ▶ Zeeschuimer → body, image(base64)
- ▶ 825 posts (positive ,negative and neutral)
- ▶ Ceph → /Volumes/storage-webisstud/data-in-progress/data-teaching/projects/project-argai-argumentative-ai-social-media-ws24/nutri_score/dataset

CHALLENGES

- ▶ Difficulty in finding negative posts on Insta.
- ▶ Non availability of local images to label studio.

ANNOTATION

- ▶ Label-studio
- ▶ Body only (text), Image only ,Post (text+image)

LABELS

- ▶ Positive-Happy,Joy ,Surprise, Funny,Sarcastic.
- ▶ Negative- Scary,Suicidal,Depressive,Violence,Abusive.
- ▶ Neutral-News,Facts.

CHALLENGES

- ▶ knuckle intensive job.
- ▶ Hashtags and emojis were reconsidered.

```
<View>

  <Header value="Post Mood Based on Image" />
  <Image name="image" value="$filename"/>
  <Choices name="post_mood_PNN" toName="image" choice="single" required="true">
    <Choice value="positive"/>
    <Choice value="Negative"/>
    <Choice value="Neutral"/>
  </Choices>

  <Header value="Post Mood Based on text" />
  <Text name="text" value="$body"/>
  <Choices name="post_mood_PNN_text" toName="text" choice="single" required="true">
    <Choice value="positive"/>
    <Choice value="Negative"/>
    <Choice value="Neutral"/>
  </Choices>

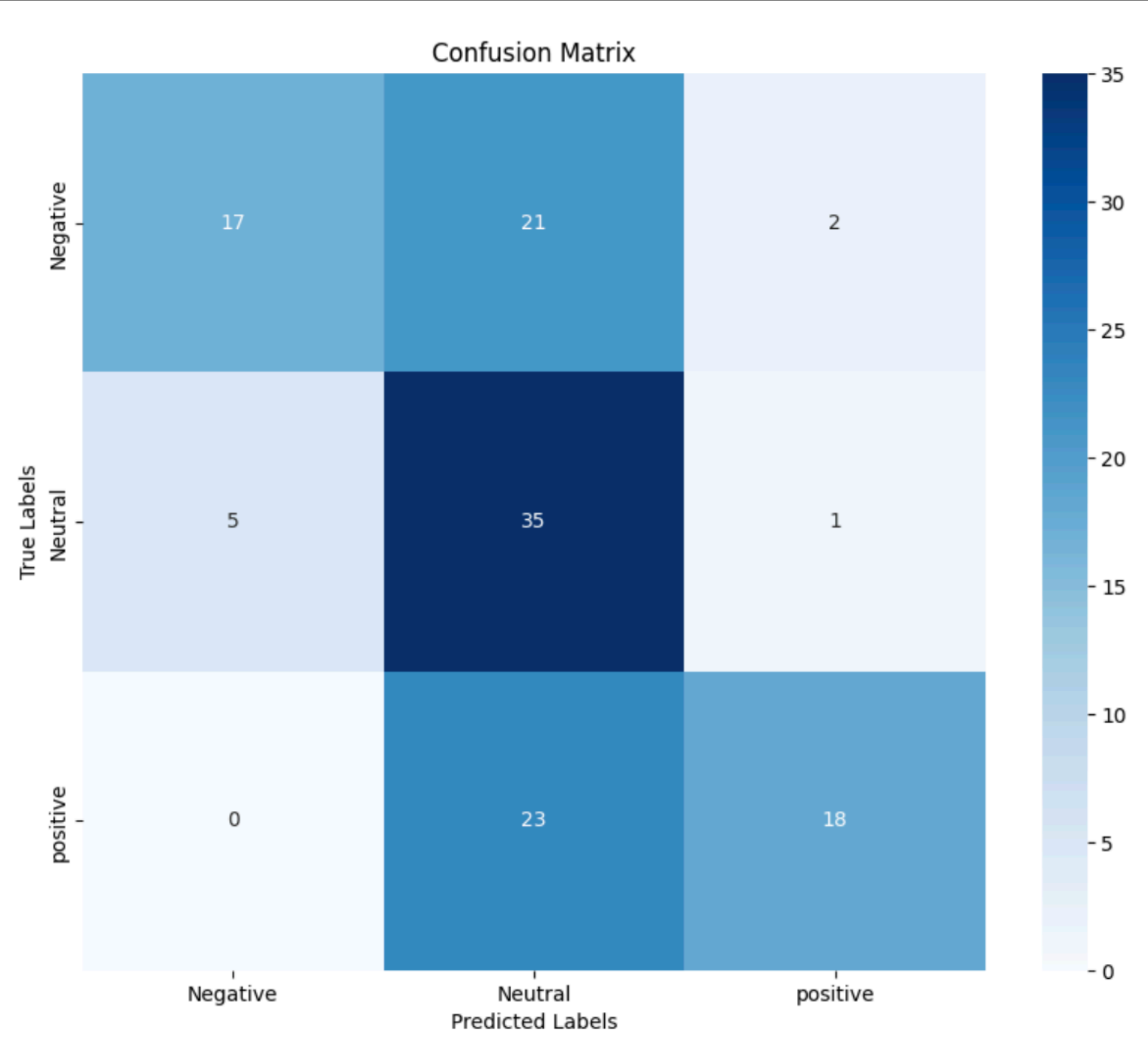
  <Header value="Post Mood Based on post" />
  <Text name="ids" value="$id"/>
  <Choices name="label_post" toName="ids" choice="single" required="true">
    <Choice value="positive"/>
    <Choice value="Negative"/>
    <Choice value="Neutral"/>
  </Choices>
</View>
```


TEXT-SENTIMENT

► Twitter-roberta sentiment model.

BASELINE

```
Metrics per class:  
Label: Negative  
  Precision: 0.7727  
  Recall: 0.4146  
  F1 Score: 0.5397  
Label: Neutral  
  Precision: 0.4430  
  Recall: 0.8537  
  F1 Score: 0.5833  
Label: positive  
  Precision: 0.8571  
  Recall: 0.4390  
  F1 Score: 0.5806  
  
Overall Accuracy: 0.5691
```



TEXT-SENTIMENT

► Twitter-roberta sentiment model.

FINETUNED:

0.7-training

0.15-validation

0.15-testing(equal number of posts for each class -41)

```
(venv_label_studio) rachanavenatiicloud.com@rachanas-MacBook-Air bin % python /Users/rachanavenatiicloud.com/venv_label_studio/bin/finetuning_roberta.py
```

Metrics per class:

Label: Negative
Precision: 0.9429
Recall: 0.8049
F1 Score: 0.8684

Label: Neutral
Precision: 0.9394
Recall: 0.7561
F1 Score: 0.8378

Label: positive
Precision: 0.6909
Recall: 0.9268
F1 Score: 0.7917

Overall Accuracy: 0.8293

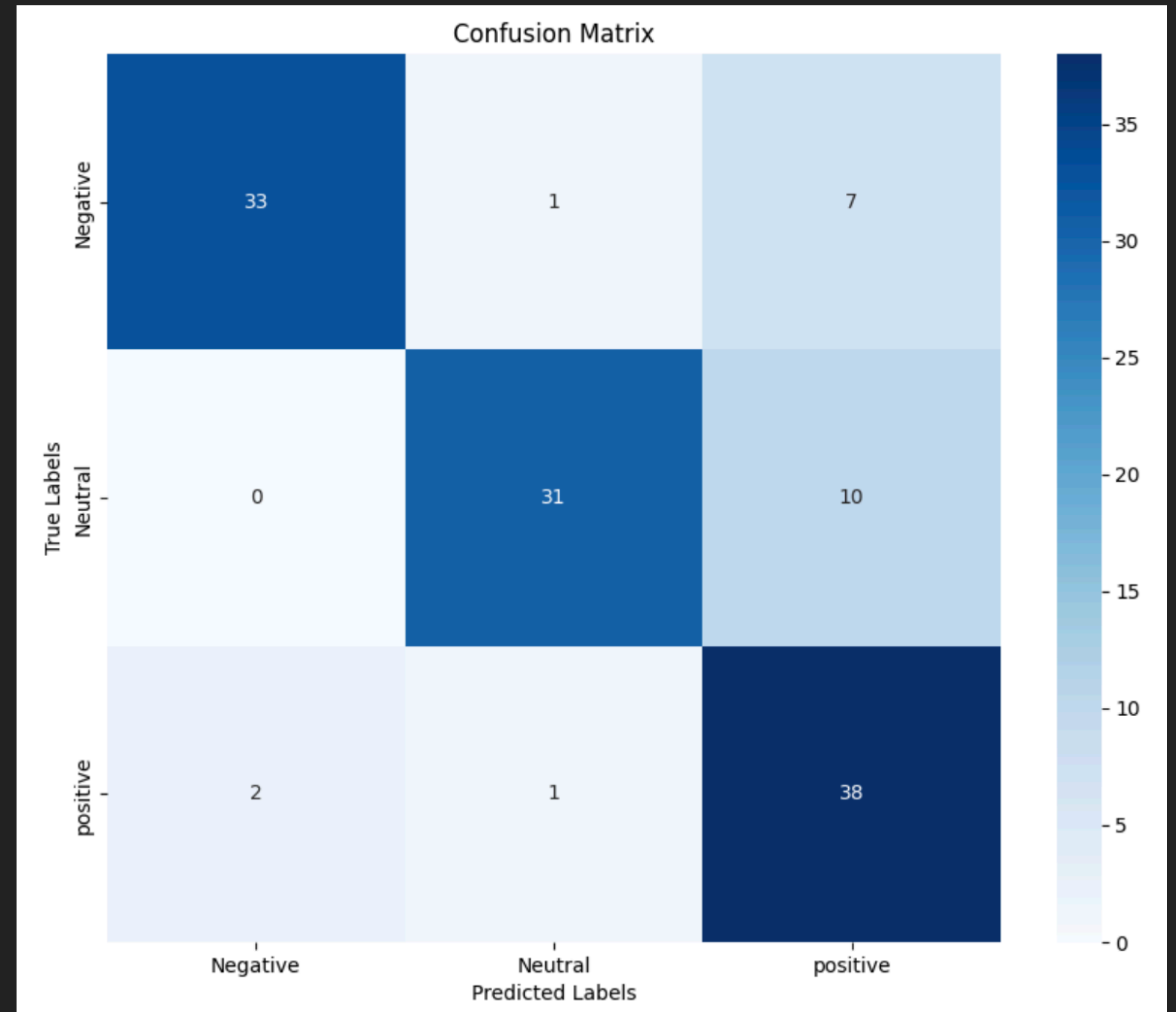


IMAGE-SENTIMENT

LLAVA

- ▶ Zero shot prompting.
- ▶ Did not fine-tune.

Label: negative

Precision: 0.4848

Recall: 0.3902

F1 Score: 0.4324

Label: neutral

Precision: 0.3231

Recall: 0.5122

F1 Score: 0.3962

Label: positive

Precision: 0.6800

Recall: 0.4146

F1 Score: 0.5152

IMAGE-SENTIMENT

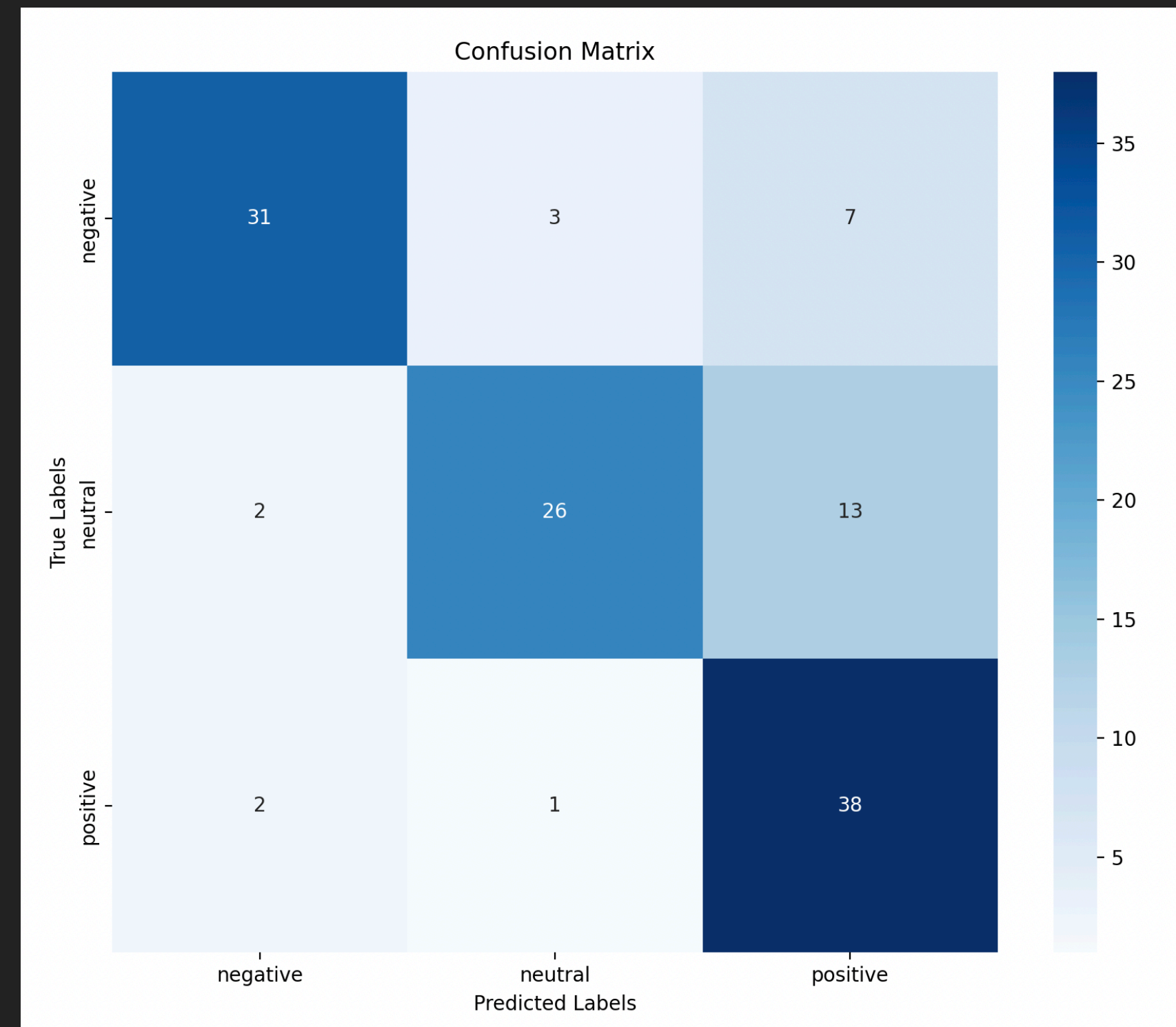
CLIP

- ▶ Struggled to converge during training.
- ▶ Loss did not decrease consistently.
- ▶ Reason: CLIP **embeddings** are **tightly clustered** making classification difficult.

IMAGE-SENTIMENT

VISUAL TRANSFORMERS

- ▶ **Classification head** (last layer) of ViT base **replaced** to **fit 3 classes**.
- ▶ Training : image $\longrightarrow$ 16 x 16 tokens $\longrightarrow$ Transformer layers $\longrightarrow$ Weights(cross entropy)



FUSION MODEL

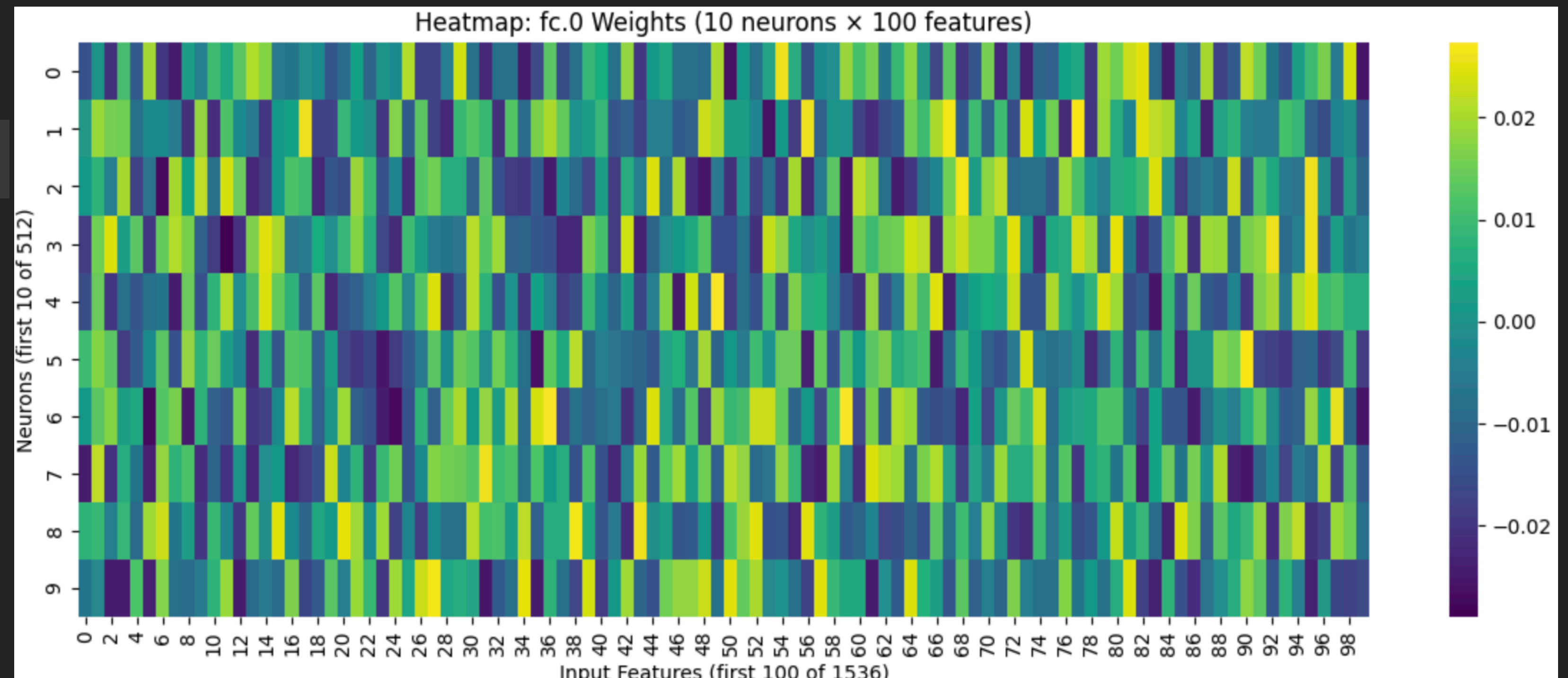
► Early fusion:

Text, Image embeddings (768 dimensions) → concatenated (1536 dimensions)

↓
Feed forward neural network (512 → ReLU → 3 logits)

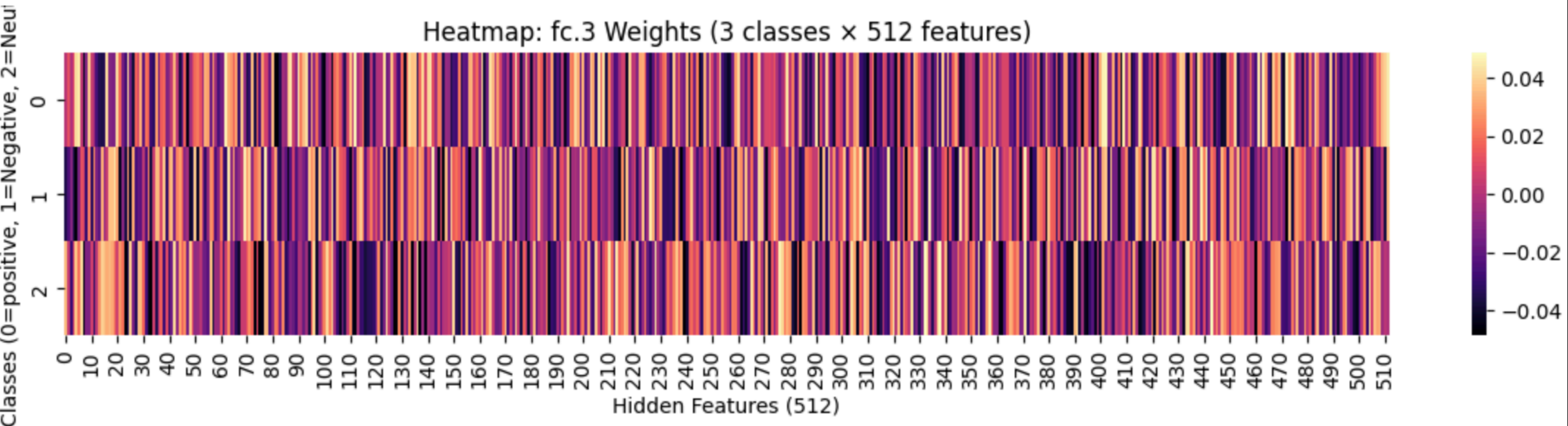
Layer: fc.0.weight
Shape: torch.Size([512, 1536])

- Many weights seem to be close to zero.
- Observe columns/features



FUSION MODEL

Layer: fc.3.weight
Shape: torch.Size([3, 512])



FUSION MODEL

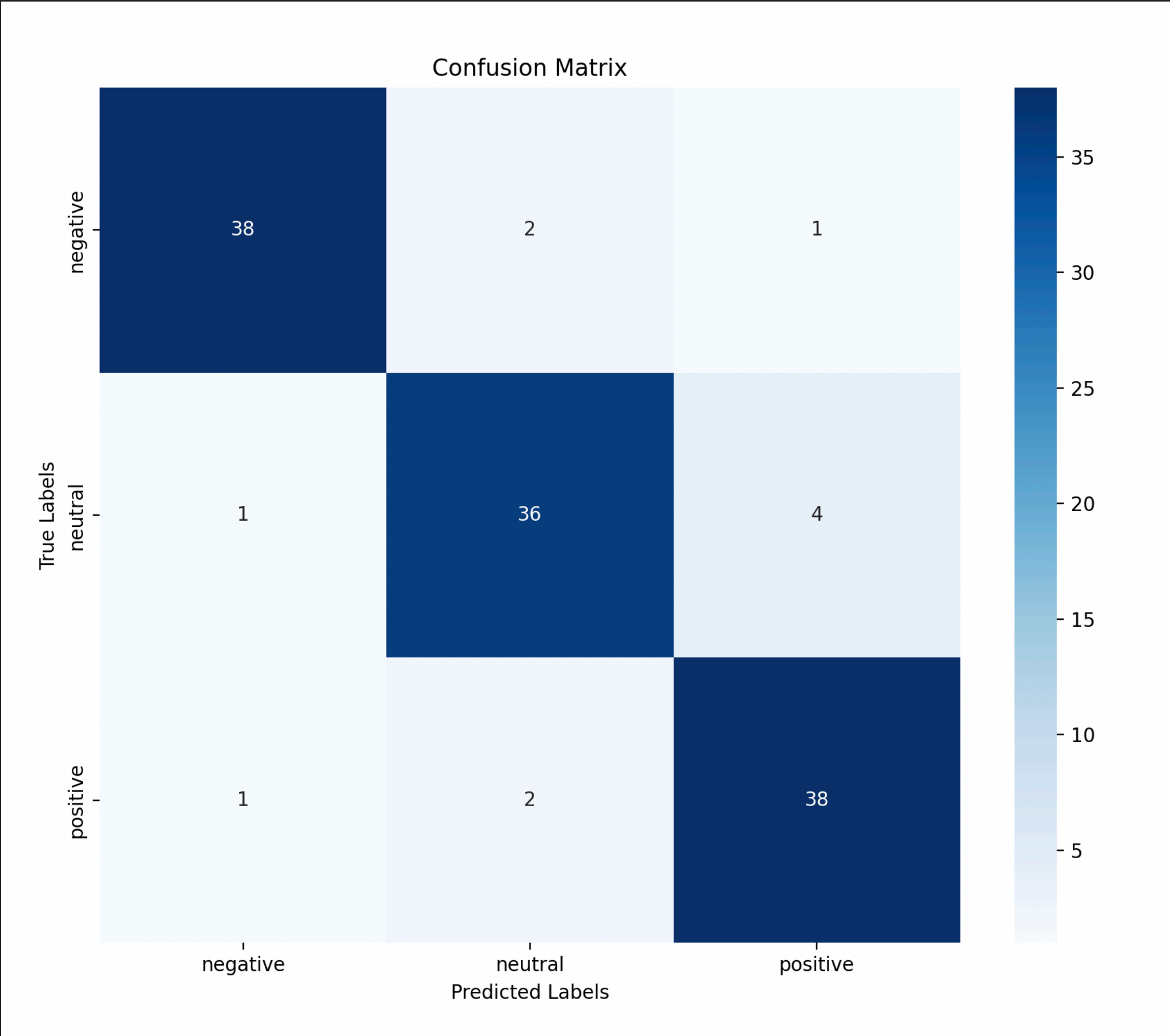
FINAL RESULTS ON TEST SET

Metrics per class:

Label: negative
Precision: 0.9500
Recall: 0.9268
F1 Score: 0.9383

Label: neutral
Precision: 0.9000
Recall: 0.8780
F1 Score: 0.8889

Label: positive
Precision: 0.8837
Recall: 0.9268
F1 Score: 0.9048



FASTAPI

- ▶ Takes a caption and image to return Sentiment of the post.
- ▶ Locally ran.

Future-

- ▶ Integrate as a web extension.
- ▶ Explore attention-based fusion ,late fusion.
- ▶ Classify sentiments to granular level.

Thank you